

Phylogenetic Analysis on the *rbcL* gene Using the DNA of Four Distantly Related Algal Species

Dataset Selection

- Tool: NCBI
- Gene: *rbcL*
- Organisms (Found using NCBI's online nucleotide search)
 - *Vaucheria litoria*: Coenocytic, Yellow-green
 - Accession: KF791053.1
 - *Macrocystis pyrifera*: Multicellular, Brown
 - Accession: KY987623.1
 - *Tetradismus obliquus*: Colonial, Green
 - Accession: MH909240.1
 - *Porphyridium purpureum*: Single-cell, Red
 - Accession: PV388837.1
- Reasoning for Gene & Organism Choice: The *rbcL* gene is a highly conserved gene found in algae. It is a widely used standard for genetic markers, DNA barcoding, and biodiversity assessments. It is important to understand the *rbcL* gene because, in algae, it codes for the main subunit (building block) of the protein Rubisco, which is the main enzyme found responsible for carbon fixation during photosynthesis. The four organisms selected (*Vaucheria litoria*, *Macrocystis pyrifera*, *Tetradismus obliquus*, & *Porphyridium purpureum*) were selected because, although they are all considered algae, they have evolved at different points in time, are from different phyla, have relatively different morphologies, and yet, they all contain the highly conserved *rbcL* gene. We will be using the DNA specifically, rather than the protein data, because DNA has the potential to reveal evolutionary changes that have occurred within the nucleotide sequences that protein sequences would otherwise hide due to their built-in redundancy (more than one codon can produce the same amino acid) (Sam Sholtis, 2022).
- Citation:
 - NIH: National Library of Medicine, National Center for Biotechnology Information
 - Sholtis, S. (2022). *Why synonymous mutations are not always silent*. Penn State University.
<https://www.psu.edu/news/eberly-college-science/story/why-synonymous-mutations-are-not-always-silent>

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BLAST Analysis

- Tool: Web BLAST, Nucleotide BLAST (Version 2.17.0)
- Species Blasting: *Vaucheria litoria*, *Macrocystis pyrifera*, *Tetrademus obliquus*, & *Porphyridium purpureum*
- Justification: The NCBI Blast program allows us to identify regions of similarity between different nucleotide sequences by comparing against a global sequence database (Morgulis et al., 2008). This comparison is then statistically analyzed against random sequences in order to identify significant matches (Morgulis et al., 2008). These matches can then be used to infer functional and evolutionary relationships, that can then be used by researchers to draw high-impact conclusions regarding organismal taxonomy and familial lineages (BLAST).
- Interpretation
 - E-Value: 0.0
 - States if matches found between sequences happened simply by chance or not. Lower the score, less the possibility it's by chance.
 - Coverage: 86%
 - Focuses on how much of the sequences line up. Higher the better the match-up.
 - Identity: 100%
 - Based on the sequence parts that line up, how many changes happened between them. The higher percentage, the better as it means there are fewer changes.
 - Because my e-value came out to 0.0, that means the matches are statistically important and did not occur by chance. 86% of my sequence successfully linked up with my target species *Vaucheria litoria*. Of that line-up, the sequences are 100% identical. This means that the *rbcL* gene is highly conserved.

BLAST[®] » blastn suite » results for RID-OTN32MAK016

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Job Title: Final Project
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Results for: 1:IdlQuery_8002516 None(437bp)

Program: BLASTN [Citation](#)

Database: core_nt [See details](#)

Query ID: IdlQuery_8002516

Description: None

Molecule type: nucleic acid

Query Length: 437

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Percent Identity: to E value: to Query Coverage: to
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	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Vaucheria litoria strain BW3 ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit gene, partial cds...	Vaucheria litoria	699	699	86%	0.0	100.00%	378	KF791053.1

- Limits: Although I can conclude that the *rbcL* gene is highly conserved amongst these algae, based on the results generated, I still can not conclude that this gene is expressed/regulated the same in all four species.
- Citation(s)
 - Zheng Zhang, Scott Schwartz, Lukas Wagner, and Webb Miller (2000), "A greedy algorithm for aligning DNA sequences", J Comput Biol 2000; 7(1-2):203-14.

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- Morgulis A, Coulouris G, Raytselis Y, Madden TL, Agarwala R, Schäffer AA. Database indexing for production MegaBLAST searches. Bioinformatics. 2008 Dec 15;24(24):2942. doi: 10.1093/bioinformatics/btn554. Epub 2008 Dec 3. Erratum for: Bioinformatics. 24:1757. PMCID: PMC2764503.
- [BLAST](#)

Multiple Sequence Alignment

- Tool: T-Coffee (Version_11.00)
- Justification: We used T-Coffee because it provides high accuracy for small datasets by using a consistency-based algorithm to align sequences.
- Observation: Although NCBI's BLAST showed a 100% identity score, T-Coffee reveals that the algae are not as closely related as one would assume using just NCBI's Blast. This is because NCBI's BLAST program compares the sequences one at a time (Wheeler & Bhagwat, 2007), whereas T-Coffee is looking at the sequences as a whole (Tommaso et al., 2011), and, as a whole, the algae show they are not related as closely as NCBI's Blast would suggest. This is an expected result because all of these algae come from their own distinct phylum, which evolved independently at different time points (Hallmann, 2016).
- Citation(s)
 - Di Tommaso P, Moretti S, Xenarios I, Orobitz M, Montanyola A, Chang JM, Taly JF, Notredame C. T-Coffee: a web server for the multiple sequence alignment of protein and RNA sequences using structural information and homology extension. *Nucleic Acids Res.* 2011 Jul;39(Web Server issue):W13-7. doi: 10.1093/nar/gkr245. Epub 2011 May 9. PMID: 21558174; PMCID: PMC3125728.
 - Moretti S, Armougom F, Wallace IM, Higgins DG, Jongeneel CV, Notredame C. The M-Coffee web server: a meta-method for computing multiple sequence alignments by combining alternative alignment methods. *Nucleic Acids Res.* 2007 Jul;35(Web Server issue):W645-8. doi: 10.1093/nar/gkm333. Epub 2007 May 25. PMID: 17526519; PMCID: PMC1933118.
 - Wallace IM, O'Sullivan O, Higgins DG, Notredame C. M-Coffee: combining multiple sequence alignment methods with T-Coffee. *Nucleic Acids Res.* 2006 Mar 23;34(6):1692-9. doi: 10.1093/nar/gkl091. PMID: 16556910; PMCID: PMC1410914.
 - Notredame C, Higgins DG, Heringa J. T-Coffee: A novel method for fast and accurate multiple sequence alignment. *J Mol Biol.* 2000 Sep 8;302(1):205-17. doi: 10.1006/jmbi.2000.4042. PMID: 10964570.
 - Wheeler D, Bhagwat M. BLAST QuickStart: Example-Driven Web-Based BLAST Tutorial. In: Bergman NH, editor. *Comparative Genomics: Volumes 1 and 2*. Totowa (NJ): Humana Press; 2007. Chapter 9. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK1734/>
 - Hallmann, A. (2016). Algae Biotechnology – green cell-factories on the rise. *Current Biotechnology*, 4(4), 389–415. <https://doi.org/10.2174/2211550105666151107001338>

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M-Coffee alignment result

MSA	
The multiple sequence alignment result as produced by T-coffee.	
T-COFFEE, Version 11.00 (Version 11.00) Cedric Notredame SCORE=269	
BAD AVG GOOD	
Vaucheria : 27 Macrocystis : 28 Tetrademus : 30 Porphyridium : 26 cons : 26	
Vaucheria Macrocystis Tetrademus Porphyridium cons	----- AACCAGGCGTAGAT--CCGTA--GAAGCTG-----CCSCTKYTGTTGC ----- GAGAGAGCGTTCTTATTCTGTATTGAAGGTACTAACCGTGTGTCAGCTTCAGGTGAAGTTAA -----
Vaucheria Macrocystis Tetrademus Porphyridium cons	----- GGGTGAATCTTCTACTGCAACATGGACGG-TAGTTGGACAGACTTATTAACTGCTTGGGATATCT -----ACGGTTACAAAGGTCG-T-----GAGCGT- AGGTCACATTATAATGTAAC-TG--CTGCTACTCAAGA-AGA--TATGTAC----- -----
Vaucheria Macrocystis Tetrademus Porphyridium cons	-----ATTGTA--ATGATTGATCTTGTAATTGGTTATAC ATAGAGCAAAAGCATATAGAGTAGATCCAGTACTGGAA--CAAGCGACCACTCTTTGCATATAT -----TGTTACGATATCAGCCGGTTCAGGTG--AAGACACCAATACATTGCTTATGT GCAGAGTT--TGCTAAGAGGTTG--GCAGTATTATGCGATTGACTTAGTAATCGTTATAC -----
Vaucheria Macrocystis Tetrademus Porphyridium cons	----- TGCTATTCA-ATCTATGTCAATTGGGCTAGAAAAGCTGATATGATTTACATTTACACCGTGCAG TGCTTATGAATGTG--ATTIATTGAAGAAGGTTCTCTTGCGAATTAAACAGCTTCTATTATT--G AGGTTACCTATCG--ACCTTTTGAAGAAGGATCAGTACTTACTTATTCACTTCAATCGTA--G TGCAATTCA-AACTATGGCTAAATGGGCAGTAAAAATGATATGATTCCTTATTACACCGTGTG -----
Vaucheria Macrocystis Tetrademus Porphyridium cons	----- GTAATTCACATATGCACGTCAAAAAATCATGGTATTAATTTCCGTGTAATTTGTAATGGATGC GTA-----ACGTTTCG-GTTTTAAAGCTGTTAAG--CTTTACGCTAGAAGACATGAGAATTC GAA-----ACCTATTTG-GTTTCAAGCTCTTCTG--CATACGTTAGAAGATTACGATCC GTAACCTACATATCTCGTCAAAAAATCACGGTATGAATTTCCGTGTAATTTGTAAGTGGATGC -----
Vaucheria Macrocystis Tetrademus Porphyridium cons	----- GTATGGCTGGTGTGATCACCATTATGCTGGTACTGTTGTAGGAAAATAGAAGGCGATCTTTAA CTTTTGTCTTACTTAAAACTTCC-AA--GGTCCAGCTACTGGTGTAT--ATTGTGGAAGAGAA CTCTGCTTACGTTAAAACTTCC-AA--GGACCTCAGATGGTATT--CAGGTGAACGTGAC GTATGGCAGGTGTGACCATATTCATGCTGCTACTGTAGTAAATTAGAAGGGGATCCTTTAA -----
Vaucheria Macrocystis Tetrademus Porphyridium cons	----- TGTTAAAGGCTTTTATAACGCTTATTATTAACATAATTAGAAATTAATCTTCTCAAGGTGTAT AGATTAGACACATTGGA--CGT----- AAATTAAACAATATGGT--CGTG--TT-----TATTAGTTGTAC TGATTAAAGGTTCTATAACACTTTACT----- -----
Vaucheria Macrocystis Tetrademus Porphyridium cons	----- TTTTTGAATGGATTGGGAGCATTACGTAACACTGTACCAAGCATCTGGTGATTAC--TG ----- AATTAAACCTAAATTAGGTTTATCAGCTAAAACTATGGACGTGCTGTTTATGAATGTTTACGTGG -----
Vaucheria Macrocystis Tetrademus Porphyridium cons	----- TGGAACAATTACATCAATTA -----A TGG-----TTTAGACTT -----A -----

Visualization of Nucleic Acids

- Tool: N•ESPrpt 0.9 (Webversion, NESPrpt.cgi 4.1.5)
- Justification: Although N•ESPrpt, can not do alignments, it is great to use alongside T-Coffee, because it evaluates the biochemical properties found in the aligned sequences in order to see if the chemical properties of the nucleotides/bases changed or stayed the same (Gouet et al., 2003).
- Observation: Based on the output, we can see that several insertion/deletion events may have occurred at the beginning and the end of the sequences, but the center blocks (block four through seven) could be considered as the conserved regions as they line up near perfectly and remain biochemically consistent across all four distantly related species.
- Citation(s):
 - Robert, X., Guillon, C. and Gouet, P. (2025) FoldScript: a web server for the efficient analysis of AI-generated 3D protein models. Nucleic Acids Res. 53(W1), W277-W282.
doi: 10.1093/nar/gkaf326 ([freely accessible online](#)).
 - Robert, X. and Gouet, P. (2014) "Deciphering key features in protein structures with the new ENDscript server". Nucl. Acids Res. 42(W1), W320-W324.
doi: 10.1093/nar/gku316 ([freely accessible online](#)).
 - Gouet, P., Robert, X. and Courcelle, E. (2003) "ESPrpt/ENDscript: extracting and rendering sequence and 3D information from atomic structures of proteins". Nucl. Acids Res. 31(13), 3320-3323.
doi: 10.1093/nar/gkg556 ([freely accessible online](#)).
 - Gouet, P. and Courcelle, E. (2002) "ENDscript: a workflow with web interface to display sequence and structure information". Bioinformatics 18(5), 767-768.
doi: 10.1093/bioinformatics/18.5.767 ([freely accessible online](#)).
 - Gouet, P., Courcelle, E., Stuart, D.I. and Metoz, F. (1999) "ESPrpt: multiple sequence alignments in PostScript". Bioinformatics 15(4), 305-308.
doi: 10.1093/bioinformatics/15.4.305 ([freely accessible online](#)).

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```
Vaucheria      .....
Macrocystis    AACCAGGCGTAGAT...CCCGTA..GAAGCTG.....CCSCTKY
Tetrademus     GAGAGAGCGTTCCTTATTCTGTATTGAAGGTACTAACCGTGCTGTTGCAGCTTCAGGTGA
Porphyridium    .....
consensus> 70  .....
```

```
Vaucheria      .....
Macrocystis    TGTTCGCGGTGAATCTTCTACTGCAACATGGACGG..TAGTTGGACAGACTTTAACTG
Tetrademus     .....ACCGTTACAAAGGTCG..T.....
Porphyridium    AGTTAAAGGTCACATATTTAAATGTAAC..TG..CTGCTACTCAAGA..AGA..TATGTAC..
consensus> 70  .....a..g..ac.....g..t.....
```

```
Vaucheria      .....
Macrocystis    CTTGCGATATCTATAGAGCAAAAGCATATAGAGTAGATCCAGTACCTGGAA..CAAGCGA
Tetrademus     .....TGTTACGATATCGAGCCGGTTCAGGTG..AAGACA
Porphyridium    .....GAGCGT..GCAGAGTT..TGCTAAAGAAGTTG..GCAGTATTATATGCATGATTGA
consensus> 70  .....a.....t..g..c..gta.....a..ga
```

```
Vaucheria      .....
Macrocystis    TCTTGTAATTCGGTTTACTGCTATTCATC..ATCTATGTCAATTTCGGCTACAAAGCTGATA
Tetrademus     CCAATTCTTTCGATATATGCTTATGAATGTG..ATTATTTCGAAGAACGTTCTCTTGCG
Porphyridium    CCAATACATTCCTTATGTAGCTTACCTATCTG..ACCTTTTCGAAGAACGATCAGTAAC
consensus> 70  cc..a..t..attG..tTATa..tGCT..tca..a..c.....ctattTCG...aaG..a....T..a..
```

```
Vaucheria      .....
Macrocystis    TGAATTTCATTTACCGTGCAAGGTAATTCTACATATGCAGTCAAATAACATGGTA
Tetrademus     AACTTATTCACTTCATTCGTA..GGA...ACGTAATTG..GTTTCAAAGCTTTCG
Porphyridium    TGAATTCTCATTTACCGTGCTGGTAATCTACATATCTGTCAAATAACATGGTA
consensus> 70  ...TT..t..ca..TT..A..cgT...GGtA...AC..TaTtc...GT...AAA...Tc..t..gt..
```

```
Vaucheria      .....
Macrocystis    TTAAATTTCGGTGAATTGTAAATGATGGGATGCGGTGGTGTGCTCACAATCATGCTG
Tetrademus     ...CITTACGTCAGAAAGACATGAGAAATTCCTTTGCTTACTTAAAACTTTC..AA..
Porphyridium    ...CATTAAGTTAGAAAGATTACGATCCTCTCTGCTTACGTTAAACATTC..AA..
consensus> 70  ...tTT..CGT..TA.....ta....G..AT..C..T..t..GCT...gt...A.....T..C.....G
```

```
Vaucheria      .....
Macrocystis    GTACTGTTGTAGCAAAATTAGAAGCCGATCCTTTAATGGTTAAAGGCTTTATATAACGTA
Tetrademus     GACCTCCACATGGTATT...CAGCTTGAAAGGTACAAATTAACAAATGGT...CGTA
Porphyridium    GTACTGTAGTAAGTAAATTAGAAGCGGATCCTTTAATGATTAAAGGTTTCATATAACGTA
consensus> 70  Gt..Ctg.....GGta..a....aagg..g....c..t..Aa..gattAAa....T..T..a....Cgt
```

```
Vaucheria      .....
Macrocystis    TATTATTAACTAAATTAGAAATTAATCTTTCTCAAGGTGTATTTTTTGAAATGGATTGGG
Tetrademus     GG...TT.....TATTAGGTGTGACAATTAAACCTAAATAGG
Porphyridium    TACT.....
consensus> 70  .....
```

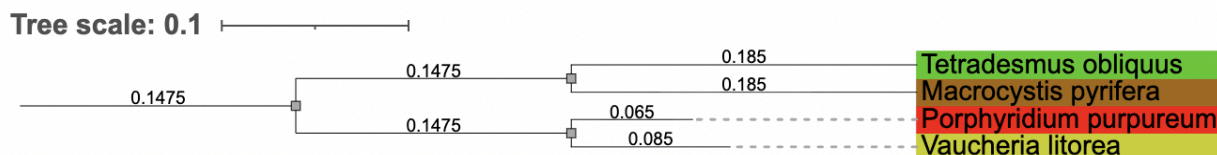
```
Vaucheria      .....
Macrocystis    CAGCATTACGTAAACTGTACCAGTAGCATCTGGTGGTATTAC..TGIGGACAATTACA
Tetrademus     TTTATCAGCTAAACTATGGACGTGCTGTTTATGAATGTTTACGTGGTGG.....TT
Porphyridium    .....
consensus> 70  .....
```

```
Vaucheria      TCAATT
Macrocystis    .....
Tetrademus     TAGACTT
Porphyridium    .....
consensus> 70  .....a
```

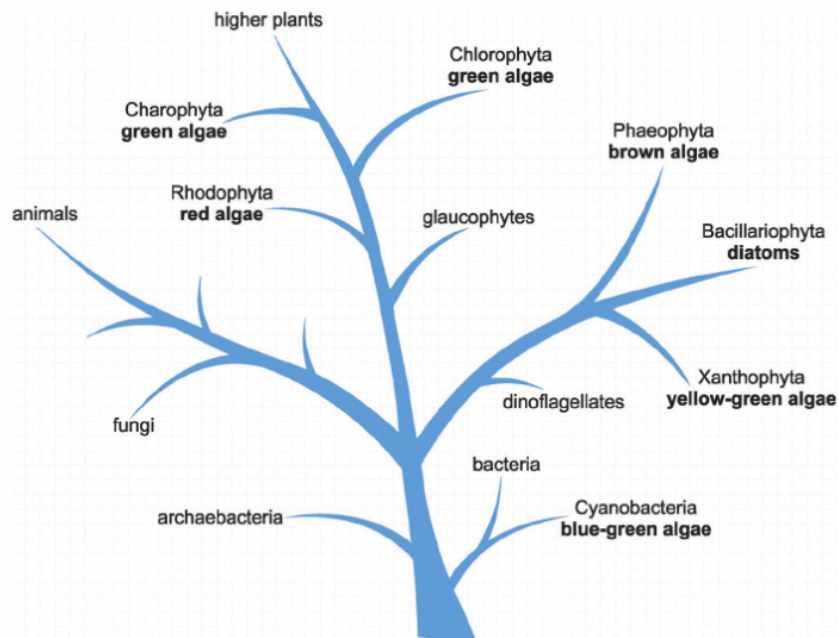
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Phylogenetic Analysis Using the DNA Alignment

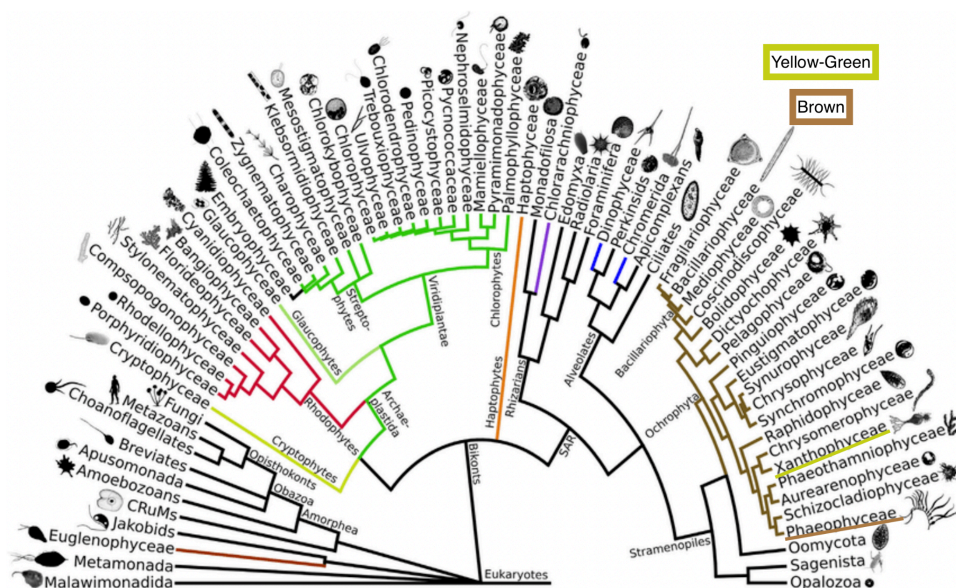
- Tool: Interactive Tree of Life (iTOL) (version 7.6)
- Justification: iTOL is a free, publicly available web program with an easy-to-use interface that effectively maps the evolutionary distance between organisms (Letunic & Bork, 2024).
- Observation: Looking at the tree, we can see that *Vaucheria litorea* is more closely related to *Porphyridium purpureum*, while *Tetrademus obliquus* is more closely related to *Macrocystis pyrifera*. This is interesting because, on a more traditional tree of life, you can see that red algae and green algae tend to be closer to each other (both belonging to the Archaeplastida group) while brown algae and yellow-green algae are more closely related (both belonging to the Heterokont/stramenopile group within the larger SAR group) (Hallmann, 2016). This organization of the tree may have occurred because of our strict focus on the *rbcL* gene. It is believed that yellow-green algae evolved from a primitive, unicellular organism that engulfed red-algae into its cellular structure (Maine Department of Environmental Protection, 2019). So, rather than digesting the red algal cell, the organelles, such as the chloroplast, became a part of the yellow-green algae (Maine Department of Environmental Protection, 2019). Then, because the chloroplast of the red algae essentially became a part of the yellow-green algae, the chloroplast of the yellow-green algae is genetically similar to that of the red algae.
- Citation(s):
 - Ivica Letunic, Peer Bork, Interactive Tree of Life (iTOL) v6: recent updates to the phylogenetic tree display and annotation tool, Nucleic Acids Research, Volume 52, Issue W1, 5 July 2024, Pages W78–W82, <https://doi.org/10.1093/nar/gkae268>
 - Hallmann, A. (2016). Algae Biotechnology – green cell-factories on the rise. *Current Biotechnology*, 4(4), 389–415. https://www.researchgate.net/publication/284206744_Algae_Biotechnology_-_Green_Cell-Factories_on_the_Rise
 - Maine Department of Environmental Protection. (2019). Algae. Biomonitoring Algae -- Monitoring and Assessment, Bureau of Land and Water Quality, Maine Department of Environmental Protection. <https://www.maine.gov/dep/water/monitoring/biomonitoring/sampling/algae/index.html>
 - Hanschen, E. R., Hovde, B. T., & Starkenburg, S. R. (2020). An evaluation of methodology to determine algal genome completeness. *Algal Research*, 51, 102019. <https://doi.org/10.1016/j.algal.2020.102019>



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[Simplified tree of life with emphasis on algae. The tree was extracted from earlier phylogenetic analyses \[1-5\].](#)



[Phylogeny of eukaryotes](#)

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Synthesis

Studying algal DNA allows us to better understand genetic conservation and organismal divergence by unmasking the effects of genetic redundancy that protein sequencing creates. Relying on a single gene (*rbcL*) can introduce single-gene bias, the oversimplification of complex traits, which in turn can create a tree that contradicts well-established organismal phylogeny. In order to mitigate this single-gene bias, phylogenetic work should focus on analyzing several conserved genes from several different organisms.